WORKED SOLUTION

STEP I 1999, Question 10

Mechanics • Simple Harmonic Motion

STEP I 1999, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q10

The key idea

The particle splits the spring into two springs, AP and PB, each with the same modulus λ as the whole spring (modulus is a property of the material) but with natural lengths a and b. Write Hooke's law for each piece in terms of a single coordinate, the distance of the particle from the upper end, and the equation of motion turns out to be linear in that coordinate. The coefficient of the coordinate gives the period, and we just look at what it does and doesn't depend on.

Solution

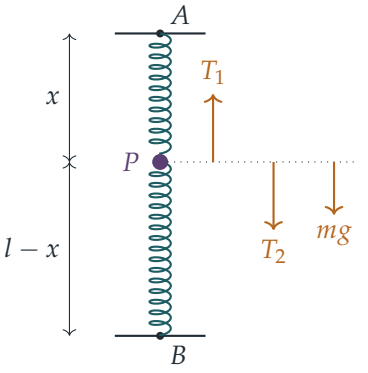
The model

Let the particle have mass m. The spring is light, so each piece has the same tension all along it, and it obeys Hooke's law in both extension and compression (it is a spring, not a string, so it can push as well as pull). Suppose first that A is the upper end. Let x be the distance of the particle below A at time t, so $AP = x$ and $PB = l - x$.

Since AP has natural length a and PB has natural length b, Hooke's law gives

$$T_1 = \frac{\lambda(x-a)}{a}, \qquad T_2 = \frac{\lambda(l-x-b)}{b},$$

where T_1 is the tension in AP and T_2 is the tension in PB. A positive tension pulls the particle towards the fixed end of that piece; a negative value means that piece is compressed and pushes the particle away. Either way the formulas give the correct signed force, so we never need to know in advance which pieces are stretched.



Simple harmonic motion

Resolving downwards for the particle, the forces are its weight mg, the tension T_1 upwards and the tension T_2 downwards:

$$\begin{aligned} m\ddot{x} &= mg - \frac{\lambda(x-a)}{a} + \frac{\lambda(l-x-b)}{b} \\ &= mg + \lambda - \frac{\lambda x}{a} + \frac{\lambda l}{b} - \frac{\lambda x}{b} - \lambda \\ &= mg + \frac{\lambda l}{b} - \lambda\left(\frac{1}{a} + \frac{1}{b}\right)x. \end{aligned}$$

The constants $\pm\lambda$ cancel, and the right-hand side is a linear function of x with negative gradient. Write

$$\omega^2 = \frac{\lambda}{m}\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{\lambda(a+b)}{mab}, \qquad x_0 = \frac{mg + \lambda l/b}{\lambda(a+b)/(ab)} = \frac{a(mgb + \lambda l)}{\lambda(a+b)}.$$

Then the equation of motion becomes

$$\ddot{x} = -\omega^2(x - x_0),$$

so, with $X = x - x_0$ (the displacement from the equilibrium position x_0), $\ddot{X} = -\omega^2 X$. This is the equation of simple harmonic motion about $x = x_0$, with period

$$\boxed{\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{mab}{\lambda(a+b)}}}$$

This proves the first result.

Ask yourself Doesn't the motion stop being simple harmonic once one piece is compressed?

It would for an elastic *string*, whose tension formula only holds while it is stretched. A spring obeys the same Hooke's law formula when it is compressed, so the single equation above holds throughout the motion (provided the coils don't close up), and there is no need to restrict l or the amplitude.

Independence of l and of orientation

The length l appears only in x_0: changing l moves the centre of the oscillation but leaves ω, and so the period, unchanged.

If instead B is uppermost, the same working applies with the roles of the two pieces exchanged: the upper piece now has natural length b and the lower one a. That just swaps a and b throughout. The period depends on a and b only through ab and $a+b$, both of which are symmetric, so it is unchanged. (The equilibrium position does change, because gravity now stretches the piece of natural length b rather than a.)

So the period is $2\pi\sqrt{mab/(\lambda(a+b))}$ whatever the value of l and whichever end is uppermost, as required.

Checks

- Two springs on either side of a mass act *in parallel*: the stiffnesses λ/a and λ/b add to give $\lambda(a+b)/(ab)$, as in ω^2 above.
- Limiting case $b \to \infty$: the lower piece becomes very weak and we recover a mass on a single spring of natural length a, with period $2\pi\sqrt{ma/\lambda}$.
- Units: λ is a force, so ma/λ has units of $\mathrm{kg\,m/N} = \mathrm{s}^2$.
- Numerically, with $m = 1$, $\lambda = 50$, $a = 1$, $b = 2$, the formula gives a period of 0.72552. Stepping the full equation of motion numerically for $l = 2$, 3 and 4.5 (including a case where both pieces are compressed), with either end uppermost, gives the same period to five significant figures, and many random choices of the constants agree to within 10^{-5}.

Common slips

- Using modulus $\lambda a/(a+b)$ or similar for the pieces. The modulus of each piece is still λ; it is the stiffness $\lambda/(\text{natural length})$ that changes.
- Getting the sign of the lower tension wrong. A stretched PB pulls the particle *down*, towards B.
- Forgetting to argue the orientation part: you need to say that swapping the ends swaps a and b, and that the period is symmetric in them.

Beyond STEP

Splitting a system into springs and adding stiffnesses is the first step towards coupled oscillators: with several masses along the spring, the equations of motion form a linear system, and the possible frequencies are the eigenvalues of a matrix (the "normal modes"). In the limit of many tiny masses one obtains the wave equation for longitudinal waves in a spring or rod. Kleppner and Kolenkow, *An Introduction to Mechanics*,

treats oscillations and normal modes.

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